THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE UNIT TEST SET C

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Regular physical activity during adolescence helps in: 1

(a) weakening bones **(b)** causing fatigue
(c) building stamina and improving mood **(d)** reducing height

2. Who studied the structure of vitamin B12 and won a Nobel Prize in chemistry in 1964? 1

(a) Rosalind Franklin **(b)** Dorothy Hodgkin
(c) Lise Meitner **(d)** Marie Curie

3. Features that distinguish males from females but are not directly involved in reproduction are: 1

(a) secondary sexual characteristics **(b)** primary organs
(c) hormones **(d)** genes

4. Which item will absorb the most heat from a campfire? 1

(a) A white cloth **(b)** A shiny foil **(c)** A mirror **(d)** A black cloth

5. The continuous movement of water between land, sea and atmosphere is the: 1

(a) food chain **(b)** carbon cycle **(c)** rock cycle **(d)** water cycle

6. Smoke rises from a burning fire because: 1

(a) hot air rises **(b)** smoke is heavy **(c)** air is cold **(d)** wood is light

7. An object covering unequal distances in equal intervals has: 1

(a) non-uniform motion **(b)** no motion
(c) uniform motion **(d)** zero speed

8. The SI unit of time is the: 1

(a) hour **(b)** second **(c)** day **(d)** minute

9. 36 km/h is equal to: 1

(a) 0.6 m/s **(b)** 36 m/s **(c)** 10 m/s **(d)** 129.6 m/s

10. Undigested waste is stored temporarily in the: 1

(a) oesophagus **(b)** liver **(c)** rectum **(d)** stomach

11. Insects breathe through tiny holes called: 1

(a) spiracles **(b)** alveoli **(c)** stomata **(d)** gills

12. In the breathing model, the rubber sheet represents: 1

(a) windpipe **(b)** diaphragm **(c)** nose **(d)** lungs

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 15–25 words each.

13. What is the menstrual cycle? State its usual duration. 2

14. What is a sea breeze? 2

15. Differentiate between uniform and non-uniform motion. 2

16. Why is the small intestine long and folded with villi? 2

17. What is egestion? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Explain the term ‘Adam’s apple’ and how it differs in boys and girls. 3

19. Differentiate between conduction, convection and radiation. 3

20. Convert 90 km/h into m/s and 8 m/s into km/h. 3

21. Explain gas exchange in alveoli. 3

22. How does a thermos flask keep contents hot or cold? 3

OR

Explain sea breeze and land breeze. 3

23. Describe an activity to find the time period of a simple pendulum. 3

OR

Distinguish between speed and distance. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

24. (i) Why are some materials good conductors? 5
(ii) List examples.
(iii) State uses of insulators.

OR

Case study: At the beach, winds blow from the sea during the day and in the reverse direction at night. 5
(i) Names of winds.
(ii) Reason for day.
(iii) Reason for night.
(iv) Who benefits?

25. (i) Differentiate breathing and respiration. 5
(ii) Where does respiration occur?
(iii) Why do we need it?

OR

Case study: A cow chews continuously while resting. 5
(i) What is this process?
(ii) Why?
(iii) Which stomach chamber?
(iv) How are birds different?

END OF QUESTION PAPER — Total: 50 Marks

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